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Combining Satellite and In-Situ Data for Ecosystem Monitoring

February 22, 2026

Today's Schedule

Day 1: Designing, building, and analysing in-situ sensor data in the Trishuli

09:30 – 10:00 Introduction from Kathmandu University

10:00 – 10:15 Computer Setup

10:15 – 10:30 Collecting GNSS data in the Trishuli

10:30 – 10:45 Tea/Coffee Break

10:45 – 11:15 Processing GNSS data to estimate water vapor changes

11:15 – 12:30 Practical Example – Working with GNSS RINEX data

12:30 – 13:30 Lunch Break

13:30 – 14:00 Low-cost environmental stations in the Trishuli

14:00 – 14:30 Introduction to using camera data for pixel tracking and river-velocity estimation

14:30 – 14:45 Tea/Coffee Break

14:45 – 15:45 Practical Example – Measuring river flow velocity with a camera

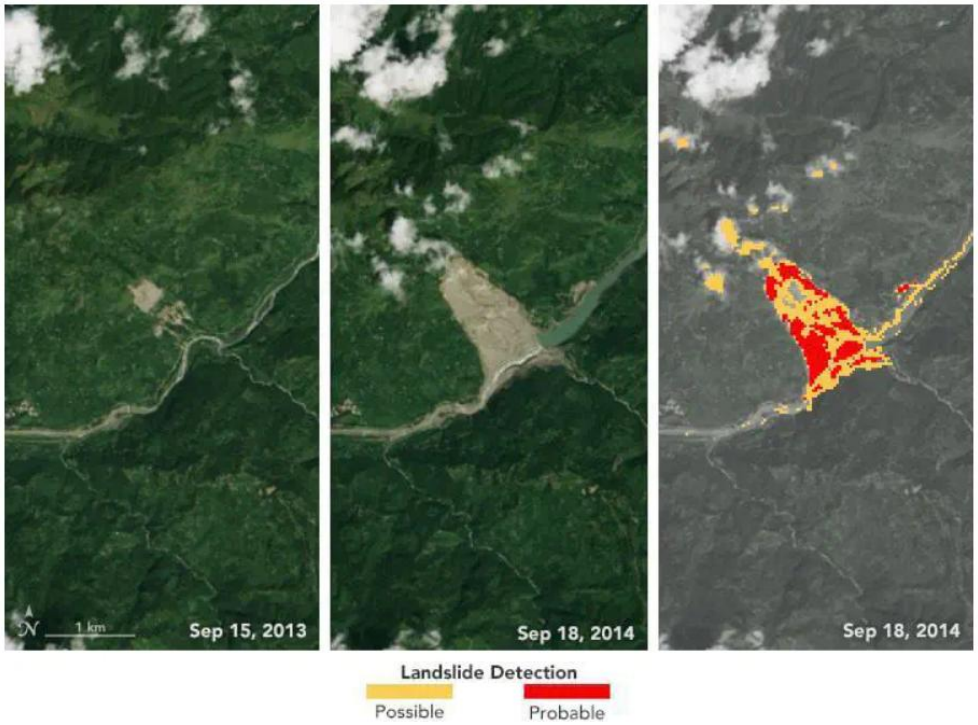
15:45 – 16:00 Wrap Up and Further Resources

Intro to Pixel Tracking

The core idea of pixel tracking is to follow how an image changes through time

This can be done in many different applications:

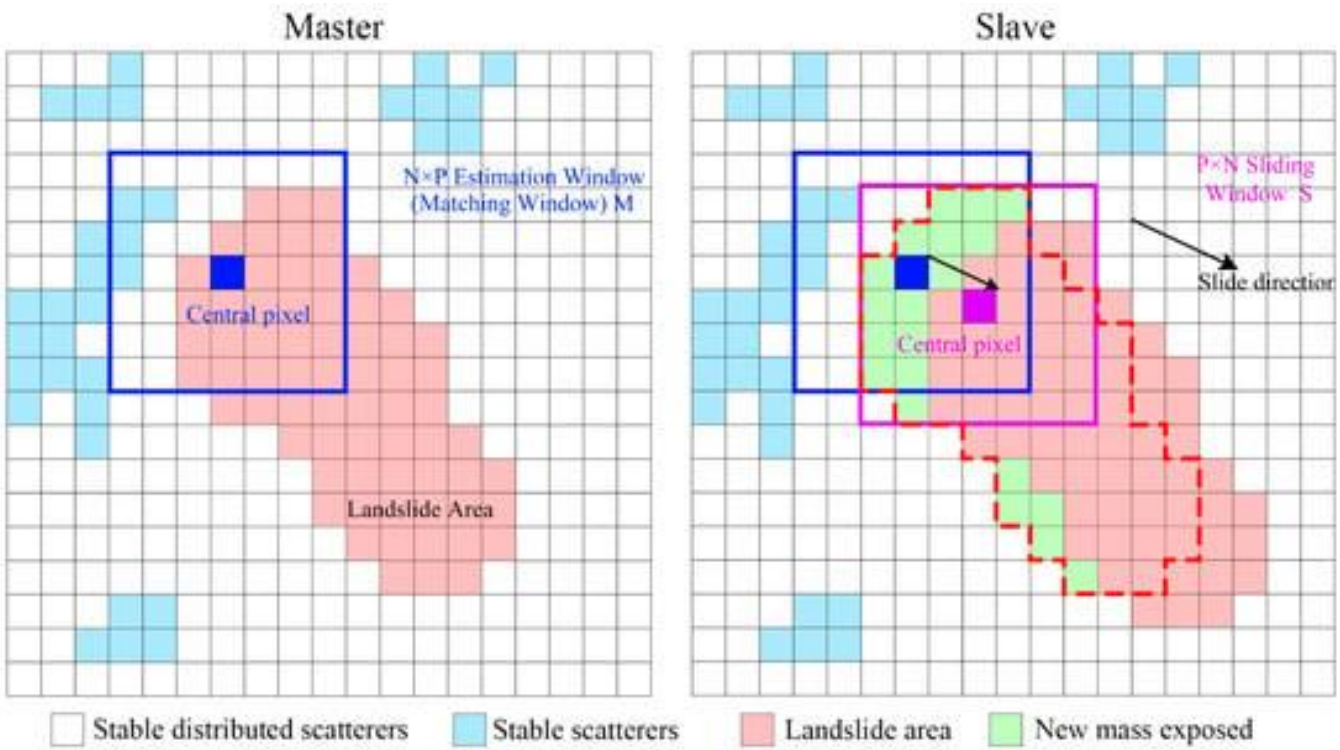
- Landslides
- Glaciers
- Rivers



There are many different algorithms to accomplish this – but they all aim to answer the same question: **How are objects in my images moving?**

https://www.waterdatalab.com/project/slip/

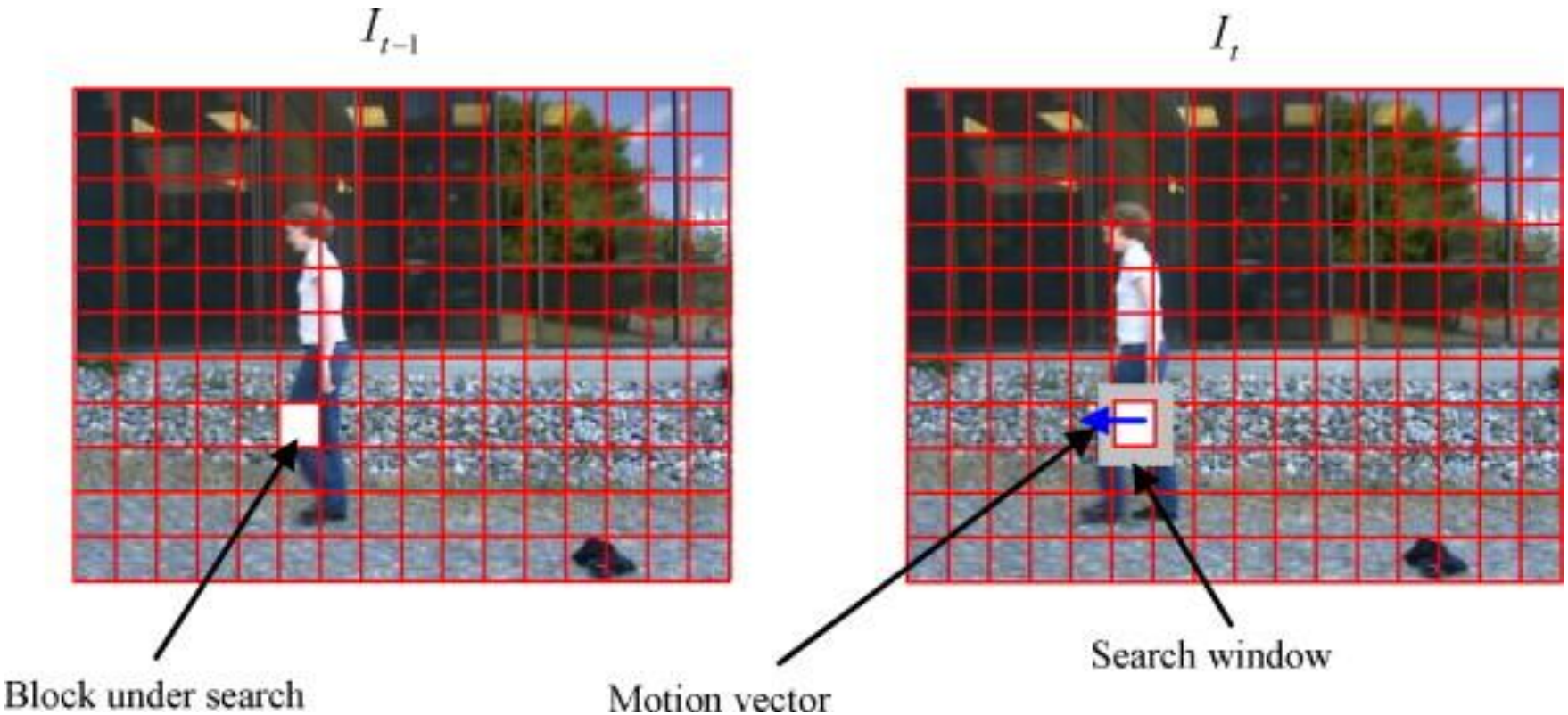
Intro to Pixel Tracking



Many algorithms work on *sliding windows* – they try to match a group of pixels from one image to the next

https://www.mdpi.com/2072-4292/17/19/3292

Intro to Pixel Tracking



https://www.sciencedirect.com/science/article/abs/pii/S1568494612004401

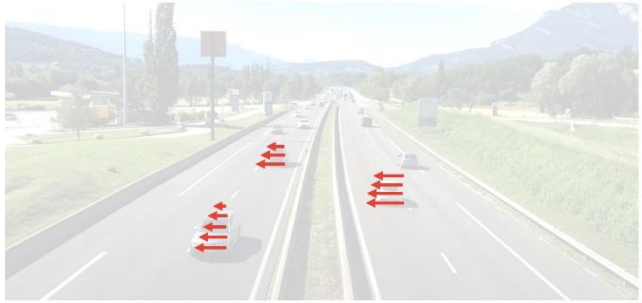
Intro to Pixel Tracking

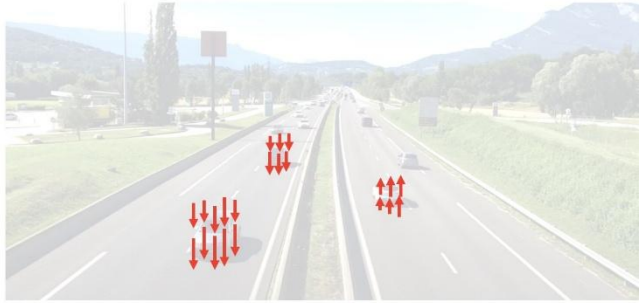
Frame t

Frame $t + dt$

$f(x, y, t)$

$f(x + dx, y + dy, t + dt)$





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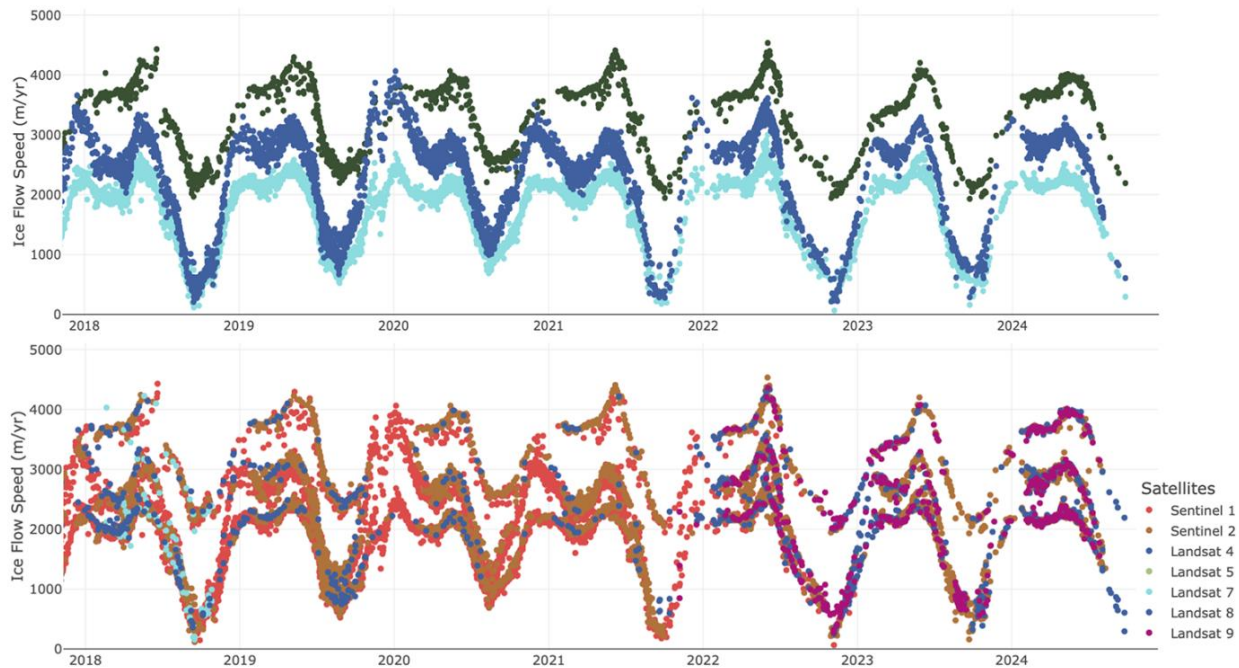
Once moving objects are identified, you can figure out how far they moved from one image to the next

Intro to Pixel Tracking



Latitude	Longitude	Symbol
60.0151	-139.4886	✖
60.0378	-139.4213	✖
60.0548	-139.3423	✖

This can be done at the point or whole-image scale, and can cover large areas!

https://tc.copernicus.org/articles/19/3517/2025/#&gid=1&pid=1

Video Data

Requirements

We need to satisfy a few requirements to be able to do this:

1. The camera must be fixed! If the camera is moving, we can’t separate the river movement from the camera movement

2. Fast data. We need to capture lots of images (or a video) in order to get short changes. If we only have images every 10 minutes, we can’t find river speed

3. The water can’t be totally flat: There has to be some ‘texture’ or other differences in the water to track

Exercise Goal: Finding Flow Velocity

Can we take a short video of a river and estimate how fast the water is moving?

How do different algorithms work?